Is it a mixedwood? Spatially-explicit responses to gradients in habitat structure and composition in three boreal bird species

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Background and Objectives

Conventional definitions of boreal mixedwood forests are based solely on tree species composition, consider species composition at a relatively coarse grain, and do not explicitly address spatial scale. Such definitions may oversimplify the habitat complexity perceived by avian species, potentially biasing past attempts to identify mixedwood habitat associations. In the absence of a clear understanding of which structural and compositional components make mixedwood forests suitable habitat for boreal birds, intimate mixedwood forests are increasingly being replaced with segregated mixedwood forests, without knowing whether these configurations provide equivalent ecological value to boreal birds. Given the ongoing decline of many boreal bird species, understanding and defining mixedwood forests from a bird's perspective is important for understanding the consequences of current harvesting and reforestation practices.

Methods

We used a boosted regression tree model to understand how the abundance of black-throated green warbler (*Setophaga virens),* Tennessee warbler (*Leiothlypis peregrina*), and bay-breasted warbler (*Setophaga castanea*) changed along a gradient of deciduous-coniferous composition, spatial configuration of conifer, and forest age within mixedwood forests. We measured environmental variables at three extents—a local (150 m), intermediate (500 m), and broad (1000 m) extent—and two grain sizes—fine (10 m² (LULC)) and coarse (30 m² (NTEMS)) grain. At each extent, we quantified forest composition as the proportion of coniferous tree species, spatial configuration of the coniferous-deciduous mixture using the Clumpiness Index (CLUMPY), and forest age as the mean forest age.

We used an existing point count dataset consisting of 8625 human and autonomous recording unit (ARU) points counts conducted between 2009 and 2024 across the Rocky Mountain, Foothills, and Boreal Forest Natural Regions of Alberta (Figure 1).

To understand how spatial configuration of the deciduous-coniferous mixture influences the abundance of our focal species, we generated predictions across synthetic landscapes representing intimate and segregated mixedwood forests at three age classes (30, 50, and 70 years). We used scale of effect predictors for each environmental variable to create a final multiscale model for predictions.

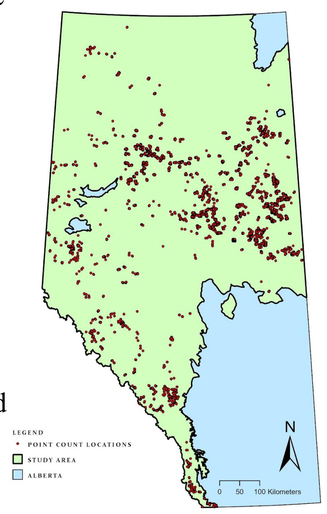

Figure 1. Locations of the point counts used in this analysis. The study area comprises the Boreal, Foothills, and Rocky Mountain Natural Regions of Alberta.

Results

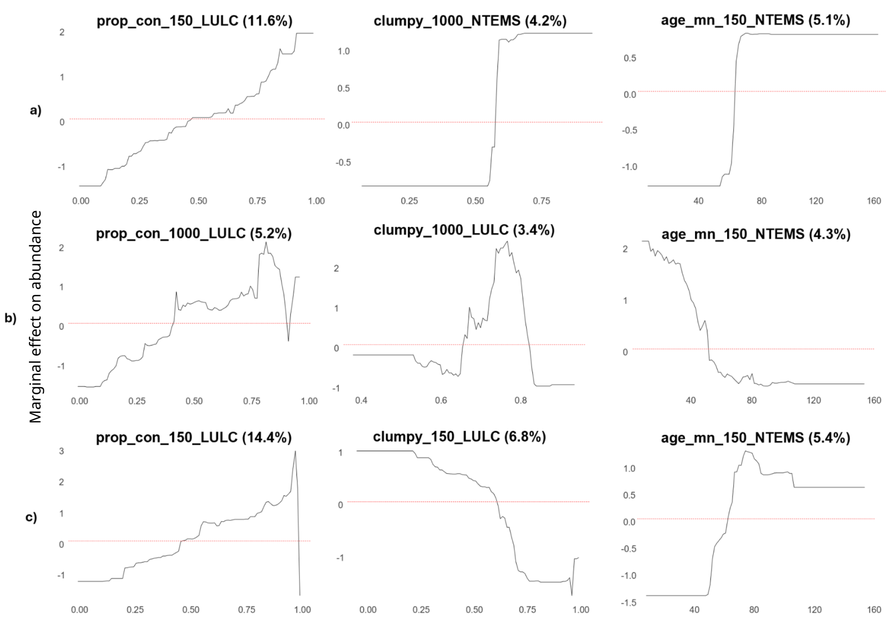

Figure 2. Partial dependence plots showing the marginal effect of each scale of effect predictor on the abundance of black-throated green warbler (a), Tennessee warbler (b), and bay-breasted warbler (c). The x-axis represents the range of values for each scale of effect predictor and the y-axis represents the centered expected count on the log scale. Red dashed lines indicate a marginal effect of zero. Numbers in parentheses show the relative contribution of each variable to the global BRT model.

The proportion of conifer emerged as the most influential predictor for all three species (Figure 3), with relative abundance increasing with the proportion of conifer (Figure 2). Spatial configuration had the second highest relative influence among scale of effect predictors for bay-breasted warbler (BBWA), and the lowest for black-throated green (BTNW) and Tennessee warbler (TEWA). Notably, BBWA responded most strongly to local extent and fine grain measurements of spatial configuration, with relative abundance decreasing with increasing aggregation. Forest age was the second most important predictor for BTNW and TEWA, and the least important predictor for BBWA. TEWA abundance decreased with increasing forest age, suggesting that TEWA prefers early-successional mixedwood forests.

Across all species, the strength and even the direction of habitat associations varied with spatial scale and grain. Finer grain sizes were generally more informative for BBWA and TEWA, whereas BTNW showed strong responses at both grain sizes depending on the predictor. The 150 m extent consistently emerged as the scale of effect across predictors. BBWA is the only species that consistently responded most strongly to local extent and fine grain size habitat measurements.

Results (continued)

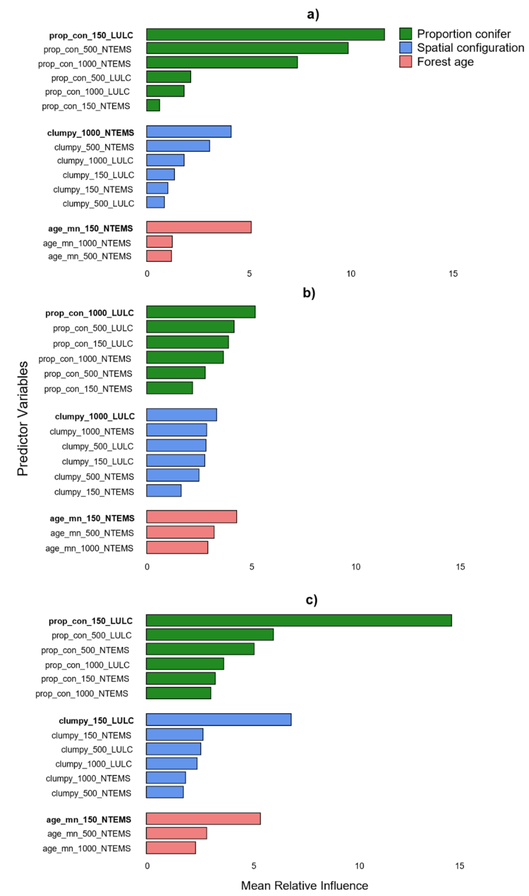

Figure 3. Relative influence (%) of predictor variables averaged across 25 bootstrapped boosted regression tree models explaining the abundance of black-throated green warbler (a), Tennessee warbler (b), and bay-breasted warbler (c). Scale of effect predictors are bolded. LULC and NTEMS represent 10 m² and 30 m² grain sizes, respectively.

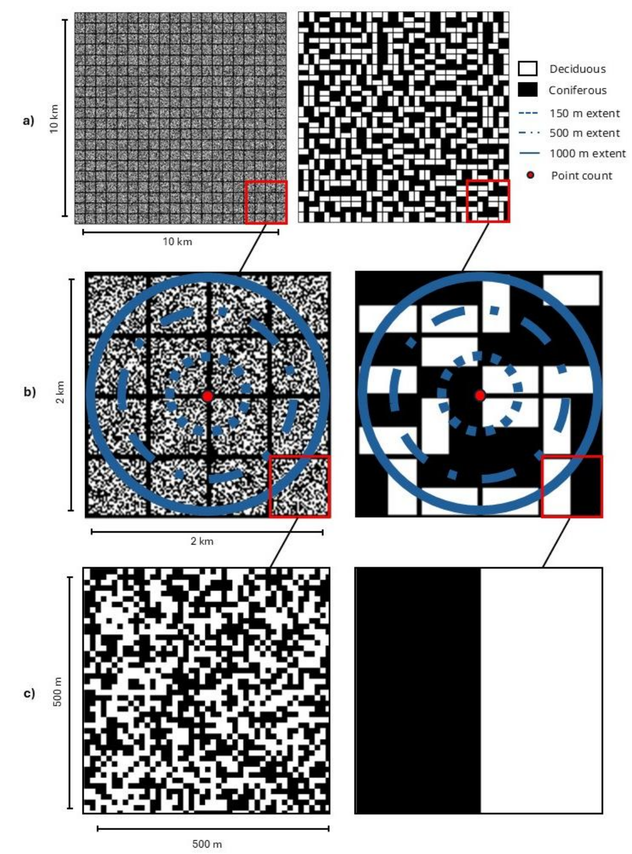

Figure 4. Synthetic intimate (left) and segregated (right) mixedwood forests used for abundance predictions. Synthetic landscapes are 10 km² (approximate size of a township) and comprise a grid of 25 ha blocks (a). A point count grid was created across synthetic landscapes for prediction. At each point count, CLUMPY and proportion conifer values were calculated at the extents and grains of scale of effect predictors (b). Conifer and deciduous pixels within synthetic landscapes were arranged with 25 ha blocks, corresponding to the average cut block size in Alberta. Conifer pixels within 25 ha blocks are 10 m² (c).

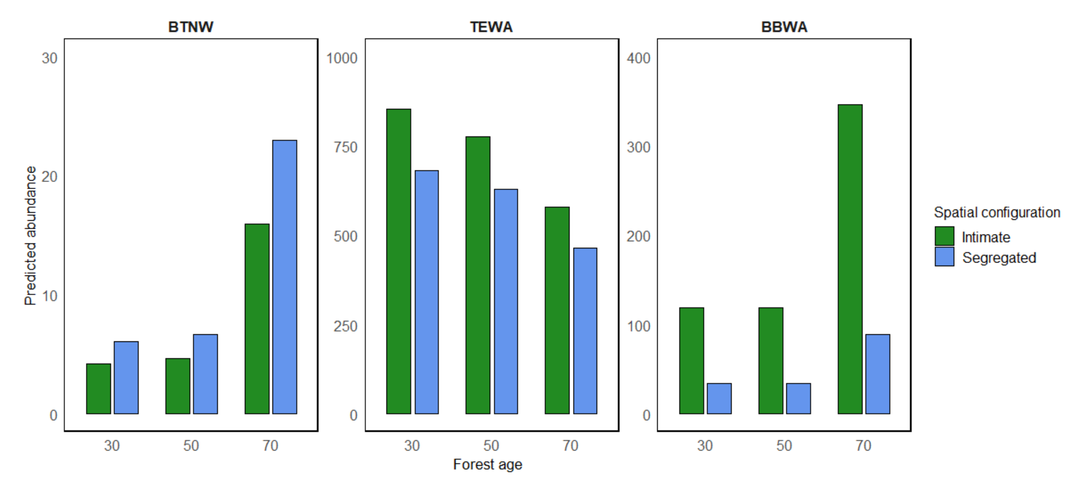

Figure 5. Predicted abundance for black-throated green warbler (BTNW), Tennessee warbler (TEWA), and bay-breasted warbler (BBWA) using a multiscale model on synthetic intimate and segregated mixedwood forests at three forest ages. Synthetic landscapes are 10 km² and contain equal proportions of coniferous and deciduous pixels. CLUMPY ≈ 0 and ≈ 1 for intimate and segregated configurations, respectively.

Our focal species showed varied responses to intimate versus segregated spatial configurations across three forest ages (Figure 5). BTNW abundance was highest in older forests (70 years) and was consistently higher in the segregated landscape across forest ages, relative to the intimate landscape. In contrast, TEWA abundance was highest in younger forests (30 and 50 years) and showed a decreasing trend with increasing forest age. TEWA abundance was higher in intimate mixedwoods than in segregated mixedwoods across all forest ages. Lastly, BBWA predicted abundance was highest in old, intimate mixedwood forests. Intimate configurations had higher predicted abundance than segregated configurations across all age classes. Differences in abundance between configurations were most pronounced in older forests, where old intimate mixedwoods had greater than 3 times the number of individuals as segregated mixedwoods, indicating a strong sensitivity to spatial configuration.

Conclusion and Implications

Our results underscore the need for species-specific, spatially explicit definitions of mixedwood forests. Relying solely on coarse classifications of stand composition may obscure important variation in how birds perceive and use habitat. Current reforestation policies in Alberta that mandate maintaining conifer proportions post-harvest, but do not address spatial configuration, risk unintentionally degrading habitat for mixedwood-associated species. Practices such as the free-to-grow standard may exacerbate this issue by encouraging conifer-dominated patches that are spatially segregated from deciduous components (Lieffers et al. 2008). Our findings suggest that preserving and restoring intimate mixtures of conifer and deciduous species at fine spatial scales may be particularly beneficial for species like bay-breasted warbler, which exhibit strong scale-dependent responses to spatial configuration.

Acknowledgements

This project is supported by funding from the Forest Resource Improvement Association of Alberta (FRIAA), Canfor, and the Natural Sciences and Engineering Research Council of Canada (NSERC).